Homework 1 — changed numbers (extra practice)

Module 1 · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A pharmacist dissolved 300 milligrams of a new antibiotic drug into a volume of purified water and placed the solution in a refrigerator (4°C). At various time intervals, the pharmacist removed a small aliquot from the solution and measured the amount of drug contained in each aliquot. The following data were obtained.

Time (hr)	Drug (mcg/mL)
0.5	226.6
1	213.9
2	190.7
4	151.5
8	95.6
12	60.4
16	38.1

1a Module 1

Is the decomposition of this antibiotic a first-order or a zero-order process?

A. Zero order

B. First order

C. Neither: it needs a third plot

1b Module 1

What is the rate constant k for the decomposition of this antibiotic?

1c Module 1

What is the half-life $t_{1/2}$?

1d Module 1

What was the initial volume of water used to prepare the original solution?

1e Module 1

How much time is required for the concentration of drug to decrease by 75%?

2. Below is the decrease in the amount of drug as a function of time.

Time (hr)	Drug A (mg)
0.5	508
1	496
2	472
3	448
6	376
9	304
12	232

Time (hr)	Drug A (mg)
16	136

2a Module 1

Does the decrease in the amount of drug A appear to be a zero-order or a first-order process?

A. Cannot tell without logarithms

B. First order

C. Zero order

2b Module 1

What is the rate constant k?

2c Module 1

What is the half-life $t_{1/2}$?

3. A solution of a drug was freshly prepared at a concentration of 280 mg/mL. After 30 days at 25°C, the drug concentration in the solution was 84 mg/mL.

3a Module 1

Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

3b Module 1

Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?